

YANG–MILLS EXISTENCE AND MASS GAP · RESEARCH PROGRAMME

Research Report 1

Locating the obstruction: the coupling window and the question of a finite reduction

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Result in brief

This report does not solve the problem, and it contains no proof of the mass gap. It answers two narrower questions.

- 1. Where exactly is the wall?** For $SU(2)$ in four dimensions with the Wilson action, rigorous mass-gap results exist only at strong coupling; the most explicit threshold we know is $\beta < 0.0833$ [SZZ23] (earlier cluster-expansion results [OSe78] also require small β). The continuum limit lives at $\beta \rightarrow \infty$, and the scaling regime begins near $\beta \approx 2.2$ (own simulation, 10 couplings, and [AT21]). Between the two lies a factor of about 26 in β , which no rigorous method currently crosses.
- 2. Could a computer close it?** Only through a *finite reduction*: a theorem that turns the infinitely many cases into finitely many certifiable inequalities. We formulate the only reduction we can identify (Proposal 3.1). It needs three new theorems, and its certification step runs into a dimensional barrier. More computing power does not remove either obstacle. A new mathematical criterion is required.
- 3. A proved limit.** The pointwise curvature (Bakry–Émery) method behind the strong-coupling result cannot, in any refinement, go beyond $\beta = 1/8$ for $SU(2)$ in $d = 4$ (Proposition 3.2, with complete proof). Local curvature arguments alone therefore provably cannot cross the window.

Status. The specification’s closure count is unchanged (1 of 9 transitions). Seven new ledger entries, A-034–A-040, record the findings.

Contents

1	Questions	2
2	Numerical laboratory (evidence class E4)	2
2.1	Set-up and validation	2
2.2	Results	3
2.3	What the numbers say, and what they do not	3
3	The reduction question	4
3.1	Why simulation cannot prove the theorem	4
3.2	How computer-assisted proofs actually work	4
3.3	A candidate reduction	5

¹Research and drafting assistance: Claude Code (Anthropic). Numerical results are evidence class E4 and literature statements were checked against the primary sources; see Section B. The AI tool is not an author [COPE23].

3.4	Obstruction 1: the coupling window	5
3.5	Obstruction 2: the dimensional barrier	6
3.6	Research questions that follow	7
4	Ledger updates	7
5	Conclusion	7
A	Deutsche Zusammenfassung	7
B	Reproducibility	8

1 Questions

The specification (`spec.pdf`) identifies the volume-uniform mass gap in physical units, ledger entry A-031, as the core open problem. Jaffe and Witten call it the place where “new ideas are needed” [JW, §6.5]. This report asks two questions about it:

- Q1** Where, quantitatively, do rigorous methods end, and where does the physics of the continuum limit begin?
- Q2** Can A-031 be reduced to a finite computation that a computer could certify? This is the only way in which computing power could decide the problem.

2 Numerical laboratory (evidence class E4)

2.1 Set-up and validation

We simulate $SU(2)$ lattice gauge theory on a periodic 12^4 lattice with the Wilson action

$$S(U) = \beta \sum_p \left(1 - \frac{1}{2} \text{Tr } U_p\right), \quad \beta = \frac{4}{g_0^2},$$

using the Kennedy–Pendleton heat bath [KP85] and two over-relaxation steps per sweep. Each coupling has 150 thermalization sweeps, 600 update sweeps and 200 measurements; errors are jackknife errors from 20 bins. The observables are the plaquette, planar Wilson loops $W(R, T)$ with $R, T \leq 4$, the Creutz ratios $\chi(R) = -\ln \frac{W(R, R)W(R-1, R-1)}{W(R, R-1)W(R-1, R)}$ [Cre80], and a zero-momentum 0^{++} glueball correlator built from APE-smearred spatial plaquettes. Its effective mass $am_{\text{eff}}(t) = \ln(C(t)/C(t+1))$ at $t = 1$ is our estimate of the lattice mass gap. The code is `lab/su2_lattice.py`.

Validation. A built-in self-test checks the quaternion algebra against 2×2 matrices, the heat-bath distribution against the exact first moment, gauge invariance and exact action conservation under over-relaxation. The full simulation reproduces the strong-coupling series $\langle \frac{1}{2} \text{Tr } U_p \rangle = u + 4u^5 + \dots$, $u = I_2(\beta)/I_1(\beta)$, within errors at $\beta = 0.5$ and $\beta = 1.0$ (Table 1), and approaches the weak-coupling form $1 - 3/(4\beta)$ at large β (Figure 1). An independent run at $\beta = 2.3$ (seed 23 instead of 11) agrees with the main run: plaquette differs by 1.6σ , $\chi(2)$ differs by 0.0σ , $am_{\text{eff}}(1)$ differs by 0.2σ (all within 2σ).

2.2 Results

Table 1 lists the measured values. Figure 1 places the plaquette against the strong- and weak-coupling expansions and marks the regime in which a mass gap is proved. Figure 2 compares the lattice mass gap with the reference data of [AT21], first in lattice units and then in physical units.

β	$\langle \frac{1}{2} \text{Tr } U_p \rangle$	$u + 4u^5$	dev. (%)	ref. [AT21]	$\chi(2)$	$\chi(3)$	$am_{\text{eff}}(1)$
0.5	0.12398(10)	0.12383	+0.12	—	1.906(347)	—	—
1.0	0.24324(9)	0.24339	−0.06	—	1.396(36)	—	—
1.5	0.36233(10)	0.36345	−0.31	—	1.009(7)	—	—
1.8	0.44088(8)	0.44001	+0.20	—	0.770(2)	0.705(140)	—
2.0	0.50142(14)	0.49410	+1.46	—	0.599(2)	0.576(31)	—
2.1	0.53462(13)	0.52217	+2.33	—	0.5029(9)	0.450(10)	—
2.2	0.56916(14)	0.55095	+3.20	—	0.4051(7)	0.316(5)	1.05(56)
2.3	0.60225(12)	0.58044	+3.62	0.60223	0.3185(4)	0.217(2)	1.05(30)
2.4	0.62999(11)	0.61061	+3.08	0.62976	0.2541(5)	0.150(1)	1.11(22)
2.5	0.65195(8)	0.64145	+1.61	0.65182	0.2139(4)	0.111(1)	0.70(10)

Table 1: Own simulation, $SU(2)$, Wilson action, 12^4 . “dev.”: relative deviation of the plaquette from the strong-coupling series. “ref.”: plaquette of [AT21], linearly interpolated inside their range $2.30 \leq \beta \leq 2.70$. A dash marks a value whose statistical error exceeds the value itself; at strong coupling the glueball correlator and the larger Wilson loops fall below the noise, as expected for heavy states and a large string tension.

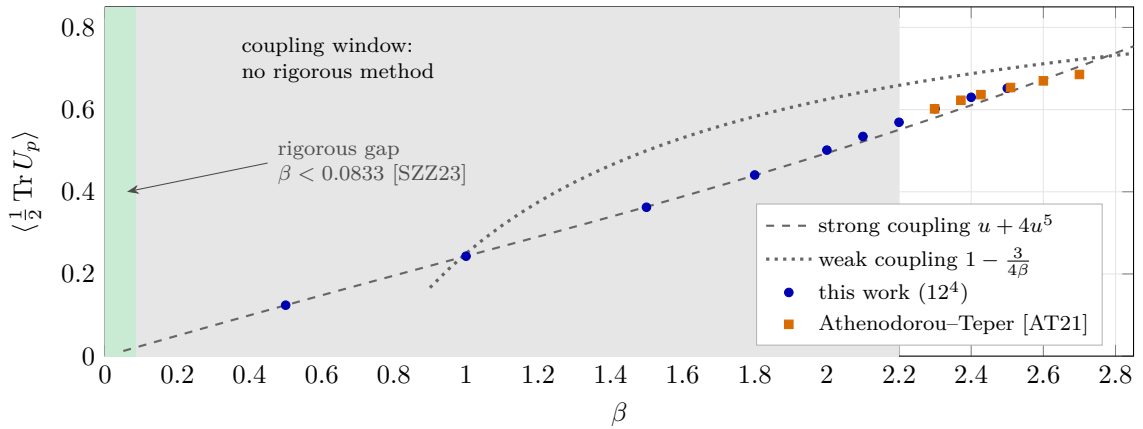


Figure 1: Plaquette against β . Green: the regime $\beta < 0.0833$ in which a mass gap is proved [SZZ23]. Grey: the coupling window up to the onset of scaling. The continuum limit is $\beta \rightarrow \infty$ to the right.

2.3 What the numbers say, and what they do not

1. **The rigorous region is tiny.** In our normalisation the Bakry–Émery / log-Sobolev argument of [SZZ23] proves a gap for $\beta < 0.0833$. Conversion: their action $S = N\beta_{\text{SZZ}} \text{Re} \sum_p \text{Tr } Q_p$ with the condition $|\beta_{\text{SZZ}}| < 1/(16(d-1))$, compared with our $\frac{\beta}{2} \text{Re} \text{Tr } U_p$ for $N = 2$, gives $\beta = 4\beta_{\text{SZZ}}$. The strong-coupling series itself describes the data numerically well beyond that point (Table 1), but this is not proved.
2. **The strong-coupling regime ends numerically near $\beta \approx 2$.** The strong-coupling series $u + 4u^5$ matches our plaquette within 0.2% up to $\beta = 1.8$ and misses it by more than 1%

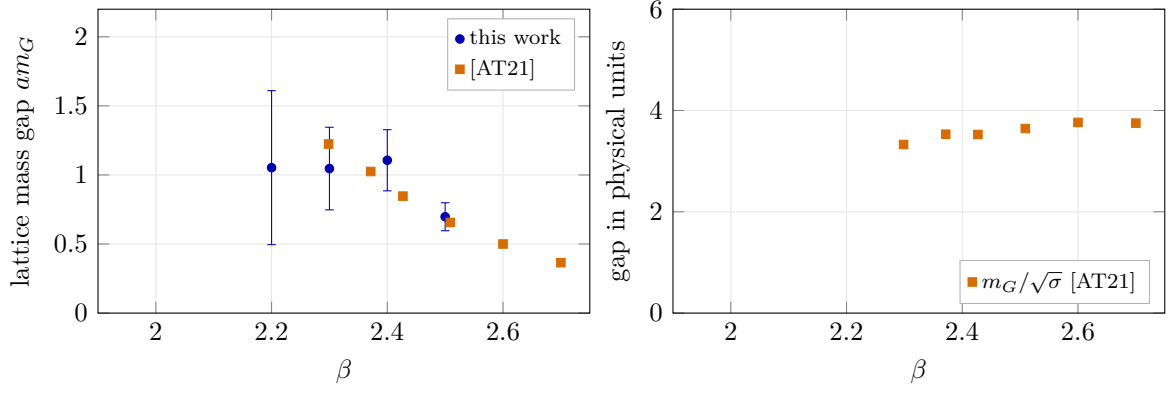


Figure 2: Left: the mass gap in lattice units falls towards zero as β grows (it must, since $a \rightarrow 0$); our estimates agree with [AT21] within their large errors. Right: in units of the string tension it stays roughly constant (reference data only; our statistics do not determine σ precisely enough). This is the numerical picture of A-031 and of rule Q-3; it is evidence, not proof.

from $\beta = 2.0$ on (Table 1). This numerical crossover lies far above the rigorous threshold $\beta < 0.0833$.

3. **Our data agree with the literature.** Inside the reference range our plaquette deviates from the interpolated values of [AT21] by at most 2.3×10^{-4} , i.e. at most 2.1σ of our statistical error, before accounting for linear interpolation and our smaller volume (Table 1). Our glueball masses agree with [AT21] within their errors of 15–53%. That precision is too low for an independent statement about the gap, so the physical-units statement below rests on [AT21].
4. **Scaling begins near $\beta \approx 2.2$.** In the literature data, $m_G/\sqrt{\sigma}$ changes by less than 15% between $\beta = 2.30$ and $\beta = 2.70$, while am_G falls by a factor of 3.4 ([AT21], Table 1; Figure 2).
5. **The window.** What is missing, in numbers, is a rigorous argument valid from $\beta \approx 0.0833$ up to the scaling regime and then along the renormalization-group trajectory to $\beta = \infty$. The coupling window alone spans a factor of about 26 in β .
6. **Limits of this evidence.** There are finitely many couplings, one volume and statistical errors. By rule Q-2, no such data can establish a bound for *all* a and L (A-012).

3 The reduction question

3.1 Why simulation cannot prove the theorem

A-031 asserts a lower bound $\Delta_{a,L} \geq c\Lambda$ for *all* sufficiently small a and large L : infinitely many cases. A simulation evaluates finitely many, with statistical errors. Even exact values for any finite list of (a, L) are compatible with $\Delta_{a,L} \rightarrow 0$ along the remaining ones. The shortcut is recorded as false in the ledger (A-012).

3.2 How computer-assisted proofs actually work

Computers have settled famous problems: the four-colour theorem [AH89], the Feigenbaum conjectures [Lan82], the Lorenz attractor [Tuc02] and the Kepler conjecture [Hal05]. All four have the same structure:

- (1) **A human theorem reduces the infinite problem to finitely many statements:** finitely many configurations, a contraction estimate on a function space, or finitely many inequalities.

- (2) **The computer certifies those statements** with rigorous error control, for example interval arithmetic. Statistical sampling is not a certificate.

For Yang–Mills, step (1) is missing. Finding it is a mathematical problem, not a computational one.

3.3 A candidate reduction

Renormalization-group (RG) block-spin transformations in the spirit of Balaban’s programme [Bal87, Bal88, Bal89a, Bal89b, Dim13a, Dim13b] map the lattice theory at spacing a to effective theories at spacings $2a, 4a, \dots$. By asymptotic freedom the effective coupling grows along the way. Stop at the first step n_* at which it reaches a fixed threshold g_* . To two-loop accuracy, $\ell\Lambda = F(g(\ell))$ with $F(g) = (b_0g^2)^{-b_1/(2b_0^2)}e^{-1/(2b_0g^2)}$. The stopping scale $\ell_* = 2^{n_*}a$ therefore satisfies

$$\ell_* \Lambda \approx F(g_*), \quad \text{independently of } a.$$

This is the structural reason why uniformity in a could come *for free*: all a -dependence goes into the number of RG steps, and what remains is a single unit-lattice problem at fixed coupling.

Proposal 3.1 (Reduction; recorded as A-039, **OPEN**). Suppose there are $g_* > 0$ and $\varepsilon > 0$ with:

- (R1) A-034 *RG step*. Uniformly in a and in the volume, the theory at spacing a is mapped to an effective unit-lattice gauge theory at spacing ℓ_* whose action lies in a class $\mathcal{A}(g_*, \varepsilon)$, with control of gauge-invariant observables.
- (R2) A-035 *Finite-volume criterion*. For effective gauge actions, a mixing condition verified on one finite box implies exponential clustering, at a rate $c > 0$, of gauge-invariant local observables in infinite volume.
- (R3) A-036 *Certification*. The condition of (R2) holds, with margin, for every action in $\mathcal{A}(g_*, \varepsilon)$.

Then A-031 holds with $\Delta_{a,L} \geq c'\Lambda$, $c' \approx c/F(g_*)$, via clustering $\Rightarrow$ gap (A-016).

Status. The implication itself is *not proved*. It is a proof sketch whose gaps include the transfer of clustering from effective to original observables. Each of (R1)–(R3) is an open theorem. The value of the proposal is that it names the missing pieces precisely.

3.4 Obstruction 1: the coupling window

(R1) is a perturbative, weak-coupling argument: it needs g_* *small*. Every currently checkable criterion for (R2)–(R3) is a strong-coupling argument: it needs g_* *large*. Examples are Dobrushin-type uniqueness conditions and the pointwise curvature condition of [SZZ23]. The reduction closes only if the two ranges overlap. For $SU(2)$ with the Wilson action, the known strong-coupling range ends at $\beta = 0.0833$, while weak-coupling behaviour sets in near $\beta \approx 2.2$ (Section 2). Balaban’s constants are not explicit, so the RG side has no number yet (A-034). The window is the precise quantitative content of “new ideas are needed”. There are only two ways to close it: push the RG through the crossover, or push a certifiable criterion up to it.

For the pointwise curvature method, the second option can be ruled out.

Proposition 3.2 (Limit of pointwise curvature methods; A-040). *Let $G = SU(N)$ with N even, $d \geq 2$, and let $\Lambda_L = (\mathbb{Z}/L\mathbb{Z})^d$ with L even. Give G^E the product of the bi-invariant metrics $\langle X, Y \rangle = -\text{Tr}(XY)$, and let $S(Q) = N\beta \text{Re} \sum_p \text{Tr} Q_p$, the sum running over plaquettes counted once. If $\beta \geq 1/(8d)$, then no $K > 0$ satisfies $\text{Ric} - \text{Hess } S \geq K$ on all of G^E .*

Proof. (i) For a bi-invariant metric, $\text{Ric} = -\frac{1}{4}B$ with the Killing form $B(X, Y) = 2N \text{Tr}(XY)$ on $\mathfrak{su}(N)$. Hence $\text{Ric} = \frac{N}{2}\langle \cdot, \cdot \rangle$ on each factor, and on the product.

(ii) Since N is even, $-1 \in SU(N)$. Put $Q_{x,\mu} = \eta_\mu(x) \mathbb{K}$ with $\eta_\mu(x) = (-1)^{x_1 + \dots + x_{\mu-1}}$, which is periodic because L is even. For $\mu < \nu$ we have $\eta_\nu(x + \hat{\mu}) = -\eta_\nu(x)$ and $\eta_\mu(x + \hat{\nu}) = \eta_\mu(x)$. Hence $Q_p = Q_{x,\mu} Q_{x+\hat{\mu},\nu} Q_{x+\hat{\nu},\mu}^{-1} Q_{x,\nu}^{-1} = -\mathbb{K}$ for every plaquette.

(iii) Right translates of one-parameter subgroups are geodesics of a bi-invariant metric. So for $X = (X_e) \in \bigoplus_e \mathfrak{su}(N)$ the curve $Q_e(t) = \exp(tX_e)Q_e$ is a geodesic with initial velocity X and $|X|^2 = \sum_e |X_e|^2$, and $\text{Hess } S(X, X) = \frac{d^2}{dt^2} S(Q(t))|_{t=0}$. All Q_e are central, so

$$Q_p(t) = Q_p e^{tX_1} e^{tX_2} e^{-tX_3} e^{-tX_4} = -\exp(tY_p + t^2 C_p + O(t^3)), \quad Y_p = X_1 + X_2 - X_3 - X_4,$$

where $X_1, \dots, X_4$ belong to the four links of p and C_p is a sum of commutators, hence traceless. Using $\text{Tr } Y_p = \text{Tr } C_p = 0$, we get $\text{Re } \text{Tr } Q_p(t) = -N + \frac{t^2}{2} |Y_p|^2 + O(t^3)$ and therefore $\text{Hess } S(X, X) = N\beta \sum_p |Y_p|^2$.

(iv) Choose $c \in \mathbb{R}^d \setminus \{0\}$ with $\sum_\mu c_\mu = 0$ and a unit vector $T \in \mathfrak{su}(N)$, and set $X_{x,\mu} = c_\mu s(x)T$ with $s(x) = (-1)^{x_1 + \dots + x_d}$. Since $s(x + \hat{\mu}) = -s(x)$, the plaquette in the plane (μ, ν) at x has $Y_p = 2s(x)(c_\mu - c_\nu)T$. Hence

$$\sum_p |Y_p|^2 = 4L^d \sum_{\mu < \nu} (c_\mu - c_\nu)^2 = 4L^d \left(d \sum_\mu c_\mu^2 - (\sum_\mu c_\mu)^2 \right) = 4d |X|^2.$$

(v) Consequently $(\text{Ric} - \text{Hess } S)(X, X) = (\frac{N}{2} - 4dN\beta) |X|^2 \leq 0$ whenever $\beta \geq 1/(8d)$. $\square$

Remark 3.3. Shen–Zhu–Zhu prove that $\text{Ric} - \text{Hess } S$ is uniformly positive for $|\beta| < 1/(16(d-1))$ [SZZ23]. The optimal threshold of the uniform curvature method therefore lies in $[1/(16(d-1)), 1/(8d)]$. For $d = 2$ both ends equal $1/16$, so their condition is sharp there. For $d = 4$ the interval is $[1/48, 1/32]$; in Wilson normalisation for $SU(2)$ it is $\beta \in [1/12, 1/8]$. That is at most 0.125 , against a scaling regime that begins near 2.2 . The value $\text{Hess } S(X, X)/(\beta |X|^2) = N \cdot 4d = 32$ was also checked by finite differences (`lab/curvature_bound.py`: 31.99996 at step 10^{-3}).

Scope. The proposition limits uniform pointwise curvature arguments only. It says nothing against log-Sobolev inequalities or mass gaps at larger β , and nothing about other methods (perturbation arguments, block-spin variables, Dobrushin-type criteria). The observation is elementary and may be known; we are not aware of it being stated explicitly.

3.5 Obstruction 2: the dimensional barrier

For spin systems, finite-volume criteria are classical: Dobrushin–Shlosman complete analyticity [DS87], and the equivalence of finite-box mixing, log-Sobolev inequalities and spectral gaps [SZ92, MO94]. They can in principle reach well beyond single-site conditions, because they are checked on a whole box. The cost is that checking them means evaluating conditional measures on that box.

For lattice gauge theory a box of side s in $d = 4$ carries about $4s^4$ link variables, each in the three-dimensional manifold $SU(2)$. That is 192 real dimensions for $s = 2$ and 3072 for $s = 4$. A deterministic certified integration has a cost that grows exponentially with the dimension, and Monte Carlo gives only statistical, not rigorous, bounds (YM-20). Only criteria whose verification is *low-dimensional* can be certified: single-site or single-block influences, or pointwise curvature bounds. Precisely these are the criteria known to hold only at strong coupling.

Consequence. Neither obstruction is lifted by more computing power, not even by many orders of magnitude. Both require a new mathematical statement. One option is a criterion that is low-dimensional to check *and* valid at intermediate coupling. The other is an RG analysis that reaches the crossover with explicit constants.

3.6 Research questions that follow

- RQ1** Is there a gauge-covariant finite-volume criterion (A-035) for lattice gauge theories? Even at strong coupling this would be a new theorem; for spin systems the analogues are classical [DS87, MO94, SZ92].
- RQ2** How far can the strong-coupling threshold $\beta < 0.0833$ be pushed for $SU(2)$, $d = 4$, with sharper curvature or block estimates? This is a well-posed and computable sub-question. It cannot close the window alone, but it measures how far local methods can reach.
- RQ3** How small must g be for Balaban’s RG estimates to hold? Making the constants explicit [Dim13a, Dim13b] would give the window its second boundary.

4 Ledger updates

The following entries are added to the assumption ledger of the specification (v1.1). No transition changes status. A-040 is our own proved statement; it carries the status **PREPRINT** until independent review.

ID	Statement	Status
A-034	(R1) RG reduction to the crossover scale, uniform in a and volume, with gauge-invariant observables	OPEN
A-035	(R2) Gauge-covariant finite-volume criterion implying infinite-volume clustering	OPEN
A-036	(R3) Certified verification of the criterion on the whole class $\mathcal{A}(g_*, \varepsilon)$	OPEN
A-037	Coupling window for $SU(2)$, $d = 4$, Wilson action: rigorous gap only at strong coupling (explicitly $\beta < 0.0833$); scaling from $\beta \approx 2.2$; am_G falls while $m_G/\sqrt{\sigma}$ stays roughly constant	NUMERICAL
A-038	Strong-coupling mass gap in infinite volume for $ \beta_{\text{SZZ}} < 1/(16(d-1))$, $SU(N)$ [SZZ23]	VERIFIED
A-039	Proposal 3.1: $(R1) \wedge (R2) \wedge (R3) \Rightarrow \text{A-031}$	OPEN
A-040	Proposition 3.2: the uniform Bakry–Émery condition fails for $\beta_{\text{SZZ}} \geq 1/(8d)$ ($SU(N)$, N even)	PREPRINT

5 Conclusion

The mass-gap problem, reduced to its computational core, is a bridge across a coupling window: from $\beta \approx 0.0833$, where local methods work, to the weak-coupling regime near $\beta \approx 2.2$, where RG methods can take over, for a whole class of effective actions and with certified bounds. Computers could take part only after a new theorem reduces this bridge to finitely many low-dimensional certified inequalities. No such theorem is known today. For the class of uniform pointwise curvature methods we prove that the bridge cannot be built: they stop at $\beta \leq 1/8$ (Proposition 3.2). This report locates the obstruction; it does not remove it.

A Deutsche Zusammenfassung

Dieser Bericht löst das Problem nicht. Er bestimmt aber genau, wo es festhängt. Für $SU(2)$ in vier Dimensionen ist ein Mass Gap bisher nur bei sehr starker Kopplung bewiesen; der explizite bekannte Schwellenwert ist $\beta < 0.0833$. Die Physik des Kontinuums beginnt erst bei $\beta \approx 2.2$. Dazwischen liegt ein Faktor von etwa 26, den keine bekannte rigorose Methode überbrückt. Eigene Simulationen und Literaturdaten zeigen: Der Gap in Gittereinheiten geht gegen null, in physikalischen Einheiten

bleibt er ungefähr konstant. Das ist genau die Aussage, die bewiesen werden müsste, hier aber nur numerisch belegt ist. Ein Computerbeweis wäre nur über eine endliche Reduktion möglich, also einen neuen Satz, der die unendlich vielen Fälle auf endlich viele prüfbare Ungleichungen zurückführt. Der einzige erkennbare Weg dorthin braucht drei neue Sätze und scheitert derzeit an zwei Hindernissen: dem Kopplungsfenster und der Dimension der zu prüfenden Integrale. Beide lassen sich nicht mit mehr Rechenleistung beseitigen, sondern nur mit neuer Mathematik. Neu bewiesen (Proposition 3.2): Die Krümmungsmethode, auf der der bekannte Beweis bei starker Kopplung beruht, kommt grundsätzlich nicht über $\beta = 1/8$ hinaus. Das Fenster lässt sich mit ihr also nachweislich nicht überbrücken.

B Reproducibility

```
cd lab
python su2_lattice.py --selftest
python su2_lattice.py --beta 2.3 --L 12 --therm 150 --meas 600 \
    --every 3 --seed 11 --out runs/beta_2.3.json      # one run per beta
python analyze.py                                     # writes reports/data/*
cd ../reports && latexmk report-01.tex
```

Raw results are in `lab/runs/*.json`. The reference values of [AT21] are copied verbatim into `lab/analyze.py`.

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